

E.3 / PAPER 1 / QUESTION 5

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5. The rest mass of an O-17 nucleus is m_{O} .

What is the binding energy of a $^{17}_8\text{O}$ nucleus?

A. $(8m_{\text{p}} + 9m_{\text{n}} - m_{\text{O}}) \times c^2$

B. $(8m_{\text{p}} + 9m_{\text{n}} + m_{\text{O}}) \times c^2$

C. $(9m_{\text{p}} + 8m_{\text{n}} + m_{\text{O}}) \times c^2$

D. $(9m_{\text{p}} + 8m_{\text{n}} - m_{\text{O}}) \times c^2$